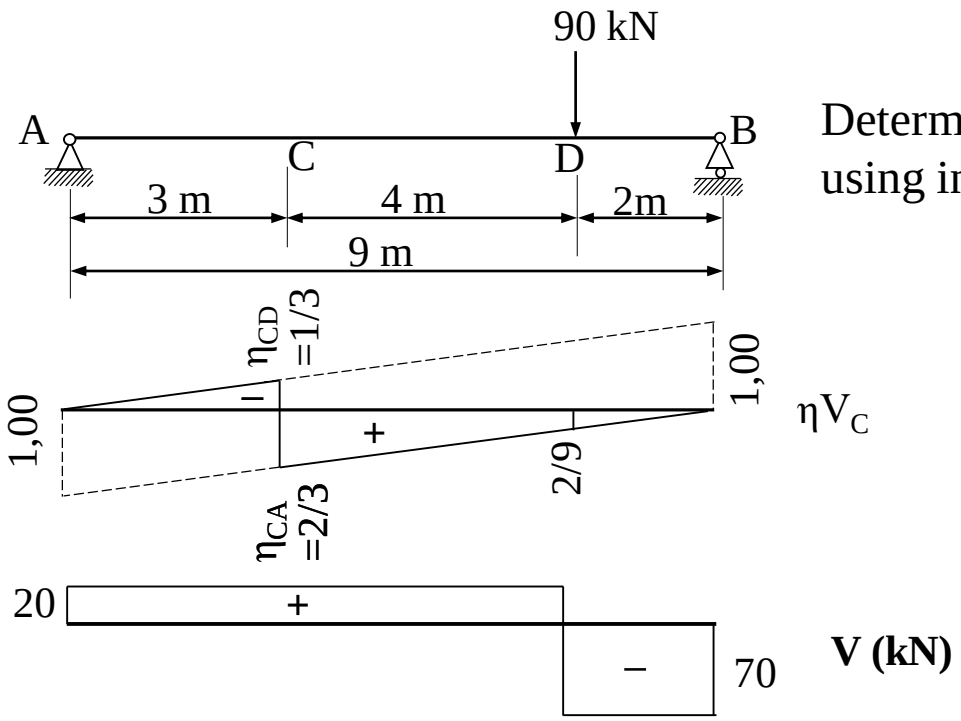
INFLUENCE LINES OF SIMPLY SUPPORTED BEAMS

Example 1)



Determine the shear force at point C by using influence line method.

Solution:

$0 < x < 3 \ \text{m} \qquad \eta V_{CD} = -x / L$

$3 \ \text{m} < x < 9 \ \text{m}$

$\eta V_{CA} = (L - x) / L$

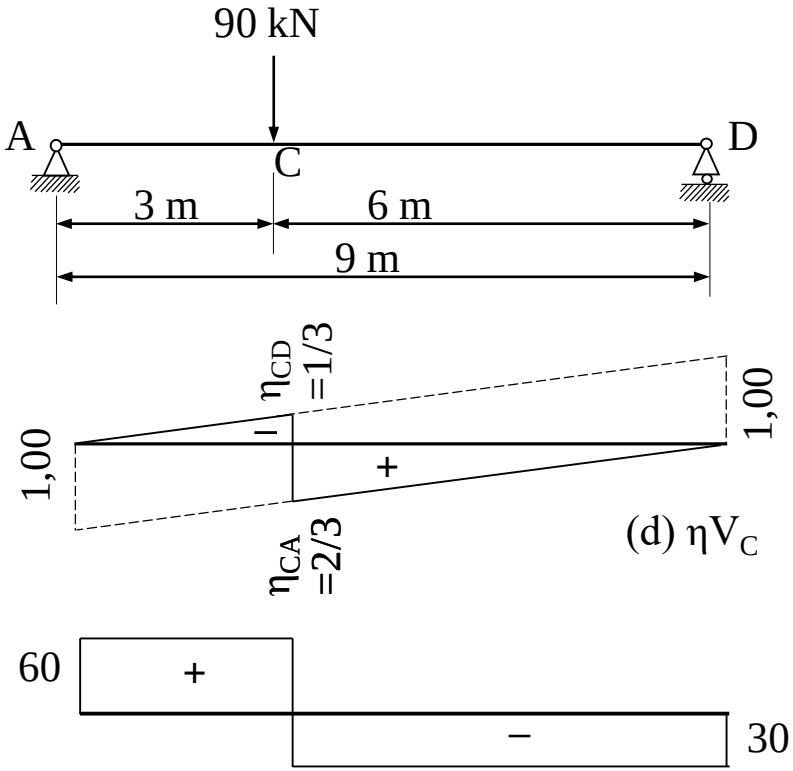
$\eta_{CA} = (6/9) \cdot 1{,}0 = 2/3$

$\eta_{CD} = -(3/9) \cdot 1{,}0 = -1/3$

$\eta_D = (2/9) \cdot 1{,}0 = 2/9$

V_C = 90x(2/9) = 20 kN

Example 2)



Determine the shear force at point C by using influence line method.

Solution:

$0 < x < 3$ m $\quad \eta V_{CD} = -x / L$

$3\ \text{m} < x < 9\ \text{m}$

$\eta V_{CA} = (L - x) / L$

$\eta_{CA} = (6/9) \cdot 1{,}0 = 2/3$

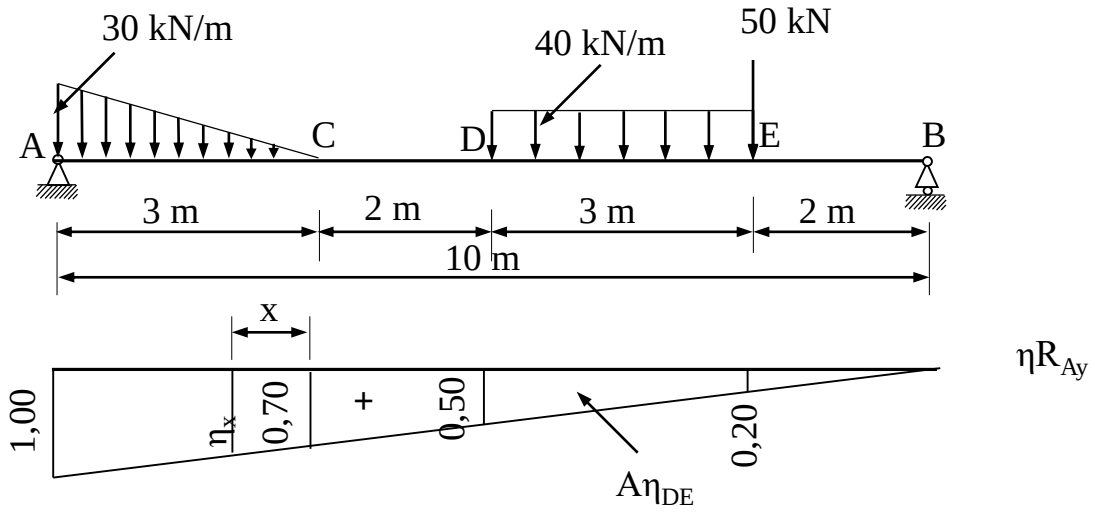
$\eta_{CD} = -(3/9) \cdot 1{,}0 = -1/3$

V_{CA} = 90x(2/3) = 60 kN

V_{CD} = 90x(-1/3) = -30 kN

Example 3)

Determine the vertical support reaction of A, shear force and moment at point C and E by using influence line method.



Solution:

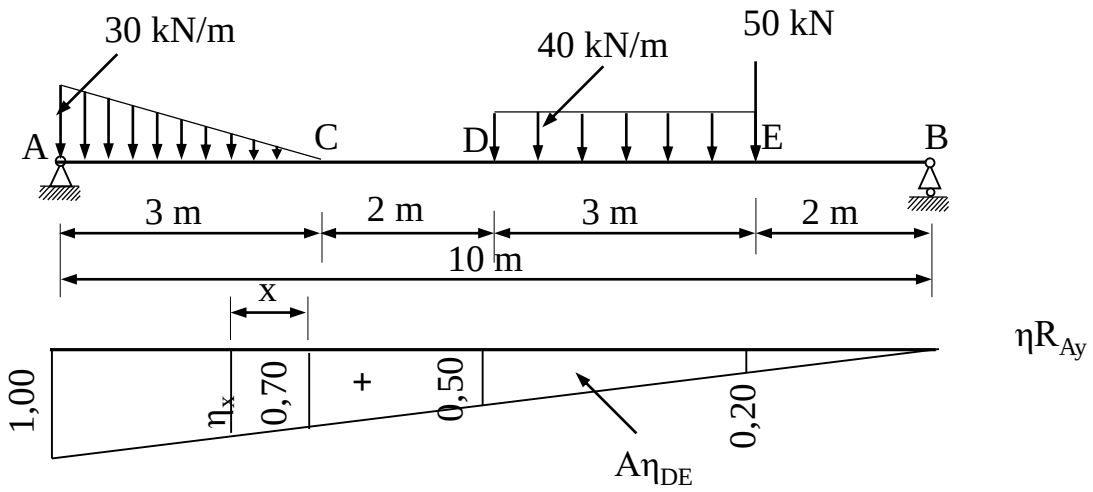
a) R_{Ay} support reaction

$\eta R_{Ay} = (L - x)/L$ x = 0 için $\eta_A = 1$ x = L için $\eta_B = 0$

$\eta_E = (2/10) \cdot 1{,}0 = 0{,}20$ $\eta_D = (5/10) \cdot 1{,}0 = 0{,}50$

$\eta_C = (7/10) \cdot 1{,}0 = 0{,}70$

$\eta_x = 0{,}70 + (1{,}0 - 0{,}70)x/3 = 0{,}70 + 0{,}10x$



$$\eta_E = (2/10)\cdot 1{,}0 = 0{,}20 \qquad \eta_D = (5/10)\cdot 1{,}0 = 0{,}50$$

$$\eta_C = (7/10)\cdot 1{,}0 = 0{,}70$$

$$\eta_x = 0{,}70 + (1{,}0 - 0{,}70)x/3 = 0{,}70 + 0{,}10x$$

At AC part $\quad q(x) = (x/3)\cdot 30 = 10x$

Influence line area of DE part:

$$A\eta_{DE} = 0{,}5\times(0{,}5+0{,}2)\times 3{,}0 = 1{,}05 \ \text{ m}$$

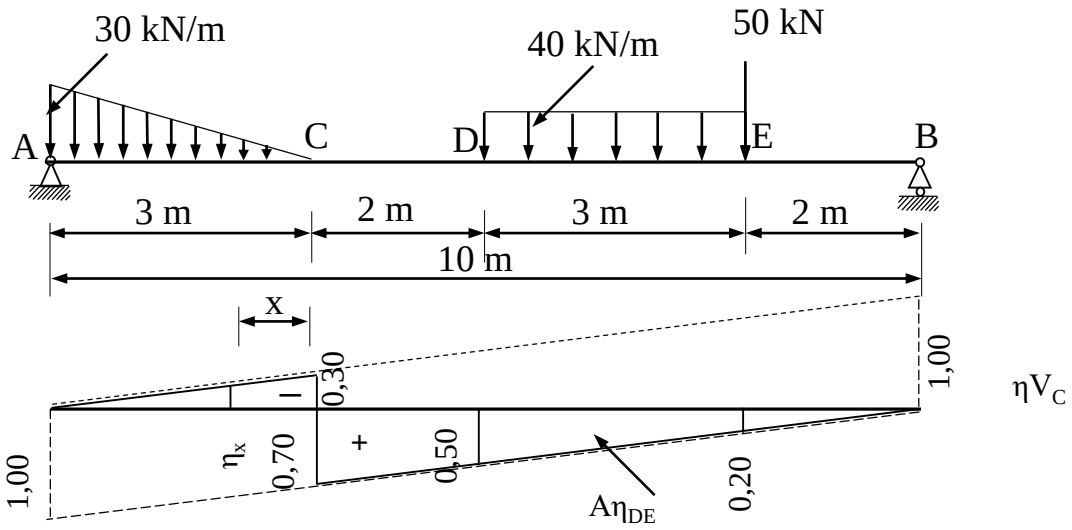
$$R_{Ay} = \int_0^3 q(x)\cdot\eta_x\cdot dx + q\cdot A\eta_{DE} + Q\cdot\eta_E$$

$$R_{Ay} = \int_0^3 10x\cdot(0{,}70+0{,}10x)dx + 40\times 1{,}05 + 50\times 0{,}20$$

$$R_{Ay} = [7x^2/2 + x^3/3]_0^3 + 42{,}0 + 10{,}0$$

$$R_{Ay} = 40{,}5 + 42{,}0 + 10{,}0 = 92{,}50 \ \text{ kN}$$

b) V_C shear force



For $0 < x < 3$ m $\quad \eta V_{CD} = -x / L$

For $3\ \text{m} < x < 10$ m $\quad \eta V_{CA} = (L - x) / L$

Substitute the boundary conditions into the equations and draw the internal force diagram of shear force of point C

$$\eta_E = (2/10) \cdot 1{,}0 = 0{,}20 \qquad \eta_D = (5/10) \cdot 1{,}0 = 0{,}50$$

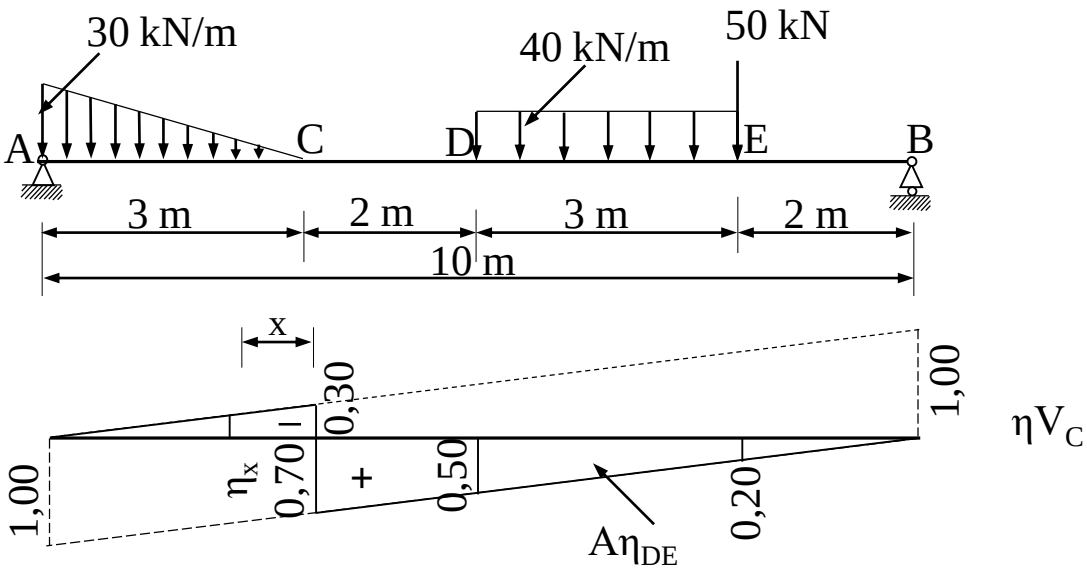
$$\eta_{CA} = (7/10) \cdot 1{,}0 = 0{,}70 \qquad \eta_{CD} = -(3/10) \cdot 1{,}0 = -0{,}30$$

$$\eta_x = -0{,}30 \cdot (3 - x) / 3 = -(0{,}30 - 0{,}10x)$$

AC part $\qquad q(x) = (x/3) \cdot 30 = 10x$

Influence line area of DE part

$$A\eta_{DE} = 0{,}5 \times (0{,}5 + 0{,}2) \times 3{,}0 = 1{,}05 \ \text{m}$$



$$q(x) = 10x$$

$$\eta_x = -(0{,}30 - 0{,}10x)$$

$$A\eta_{DE} = 1{,}05 \ \text{m}$$

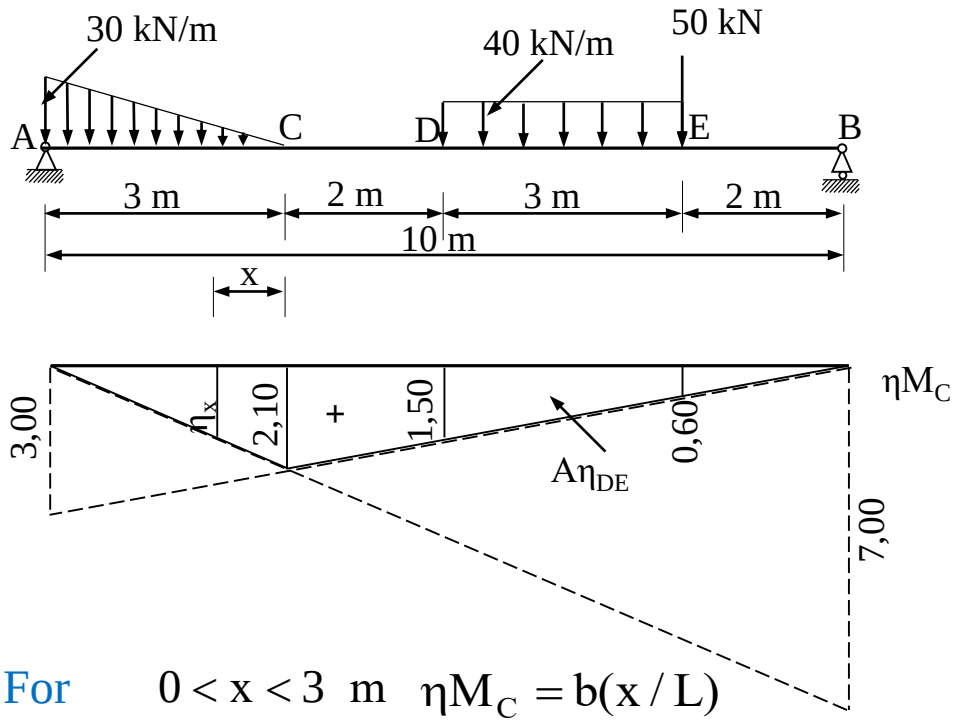
$$V_C = \int_0^3 q(x) \cdot \eta_x \cdot dx + q \cdot A\eta_{DE} + Q \cdot \eta_E$$

$$V_C = -\int_0^3 10x \cdot (0{,}30 - 0{,}10x)dx + 40 \times 1{,}05 + 50 \times 0{,}20$$

$$V_C = -[3x^2 / 2 - x^3 / 3]_0^3 + 42{,}0 + 10{,}0$$

$$V_C = -4{,}5 + 42{,}0 + 10{,}0 = 47{,}50 \ \text{kN}$$

c) M_C bending moment



For $0 < x < 3$ m $\eta M_C = b(x/L)$

For 3 m $< x < 10$ m $\eta M_C = a[(L-x)/L]$

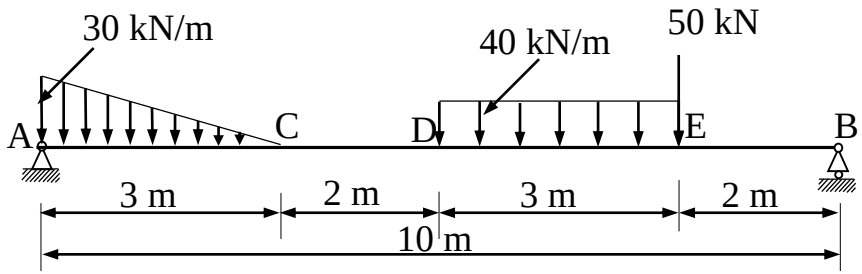
Substitute the boundary conditions into the equations and draw the internal force diagram of moment of point C

$\eta_E = (2/10)\cdot 3{,}0 = 0{,}60$ m $\quad \eta_D = (5/10)\cdot 3{,}0 = 1{,}50$ m

$\eta_C = (3/10)\cdot 7{,}0 = 2{,}1$ m $\quad \eta_x = 2{,}10\cdot(3-x)/3 = 2{,}10 - 0{,}70x$

$A\eta_{DE} = 0{,}5\times(1{,}50+0{,}6)\times 3{,}0 = 3{,}15$ m^2

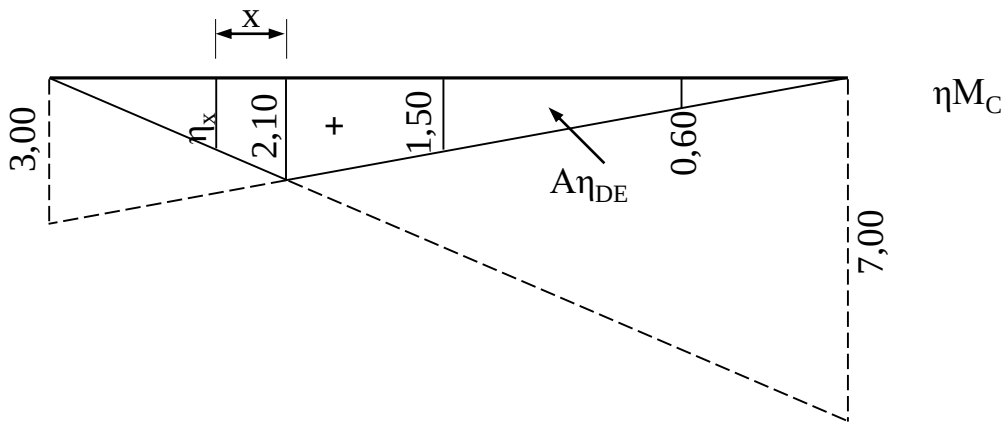
$q(x) = (x/3)\cdot 30 = 10x$



$$q(x) = 10x$$

$$\eta_x = 2{,}10 - 0{,}70x$$

$$A\eta_{DE} = 3{,}15 \ \text{m}^2$$



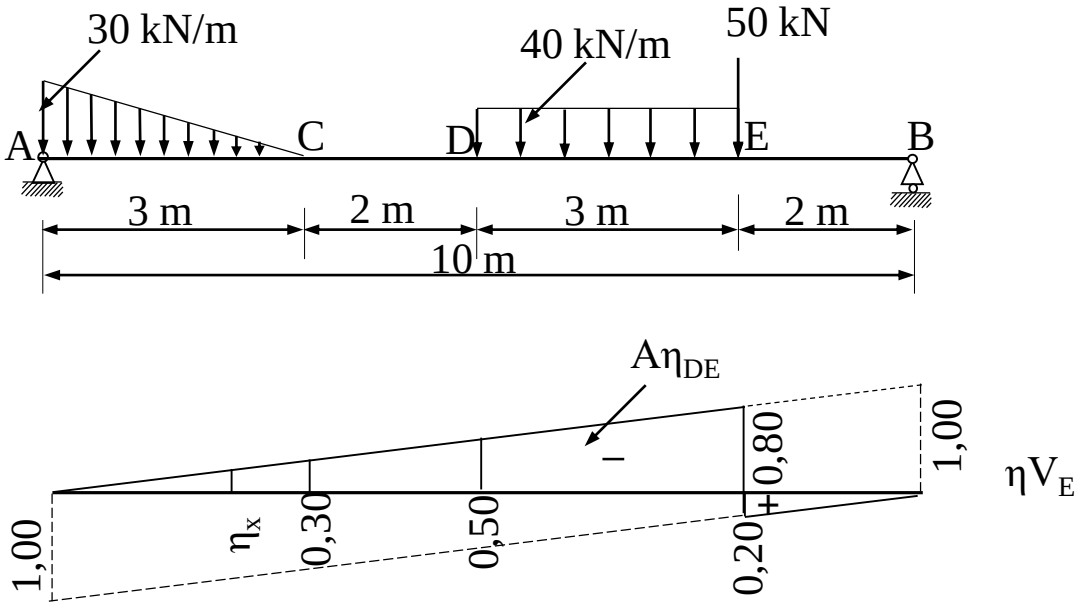
$$M_C = \int_0^3 q(x) \cdot \eta_x \cdot dx + q \cdot A\eta_{DE} + Q \cdot \eta_E$$

$$M_C = \int_0^3 10x \cdot (2{,}1 - 0{,}70x) \cdot dx + 40 \times 3{,}15 + 50 \times 0{,}60$$

$$M_C = [21 \cdot x^2 / 2 - 7x^3 / 3]_0^3 + 126{,}0 + 30{,}0$$

$$M_C = 31{,}50 + 126{,}0 + 30{,}0 = 187{,}50 \ \text{kNm}$$

d) V_E shear force



For $0 < x < 8$ m $\quad \eta V_{EB} = -x / L$

For 8 m $< x < 10$ m $\quad \eta V_{ED} = (L - x) / L$

Substitute the boundary conditions into the equations and draw the internal force diagram of shear force of point E

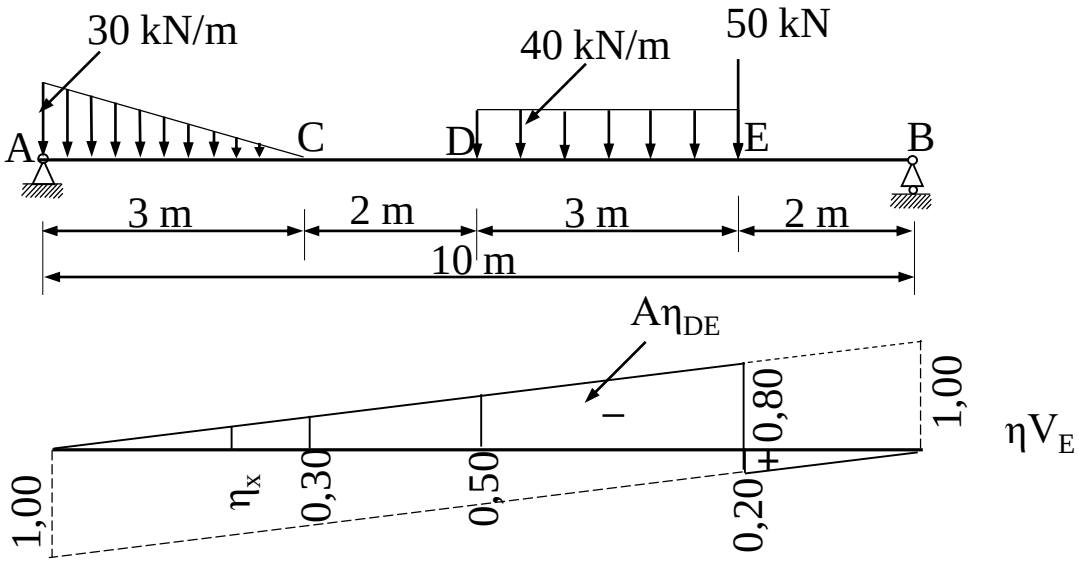
$\eta_C = -(3/10) \cdot 1{,}0 = -0{,}30 \qquad \eta_D = -(5/10) \cdot 1{,}0 = -0{,}50$

$\eta_{EB} = -(8/10) \cdot 1{,}0 = -0{,}80 \qquad \eta_{ED} = (2/10) \cdot 1{,}0 = 0{,}20$

$\eta_x = -0{,}30 \cdot (3 - x) / 3 = -(0{,}30 - 0{,}10x)$

$q(x) = (x/3) \cdot 30 = 10x$

$A\eta_{DE} = -0{,}5 \times (0{,}5 + 0{,}8) \times 3{,}0 = -1{,}95$ m



$$q(x) = 10x$$

$$\eta_x = -(0{,}30 - 0{,}10x)$$

$$A\eta_{DE} = -1{,}95 \text{ m}$$

Point load at E creates sudden change on shear force diagram. V_{ED} and V_{EB} are estimated for left and right of point E.

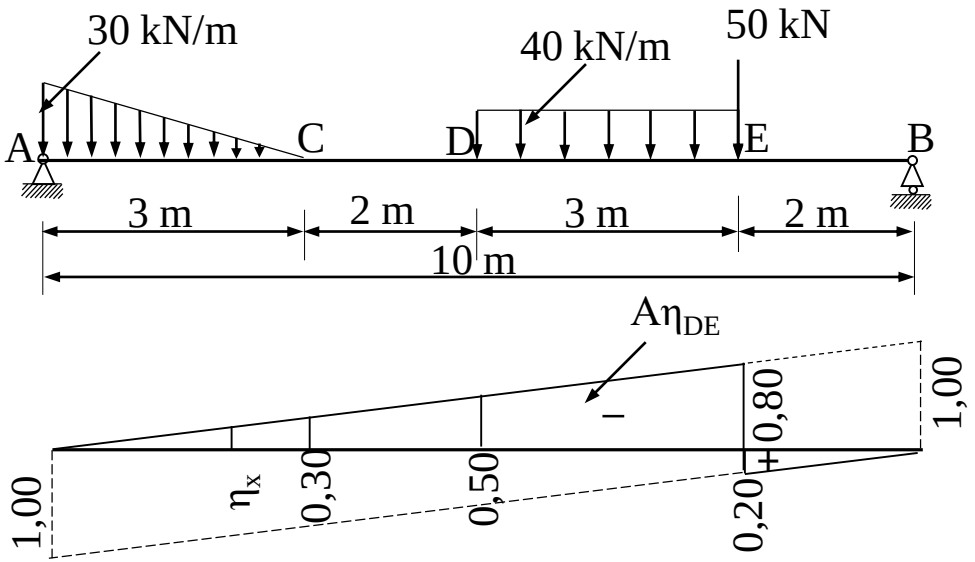
Thus, V_{ED} is calculated with the multiplication of η_{ED} and V_{EB} is calculated with the multiplication η_{EB}.

$$V_{ED} = \int_0^3 q(x) \cdot \eta_x \cdot dx + q \cdot A\eta_{DE} + Q \cdot \eta_{ED}$$

$$V_{ED} = -\int_0^3 10x \cdot (0{,}30 - 0{,}10x)dx - 40 \times 1{,}95 + 50 \times 0{,}20$$

$$V_{ED} = -[3x^2/2 - x^3/3]_0^3 - 78{,}0 + 10{,}0$$

$$V_{ED} = -4{,}5 - 78{,}0 + 10{,}0 = -72{,}50 \text{ kN}$$



$$q(x) = 10x$$

$$\eta_x = -(0{,}30 - 0{,}10x)$$

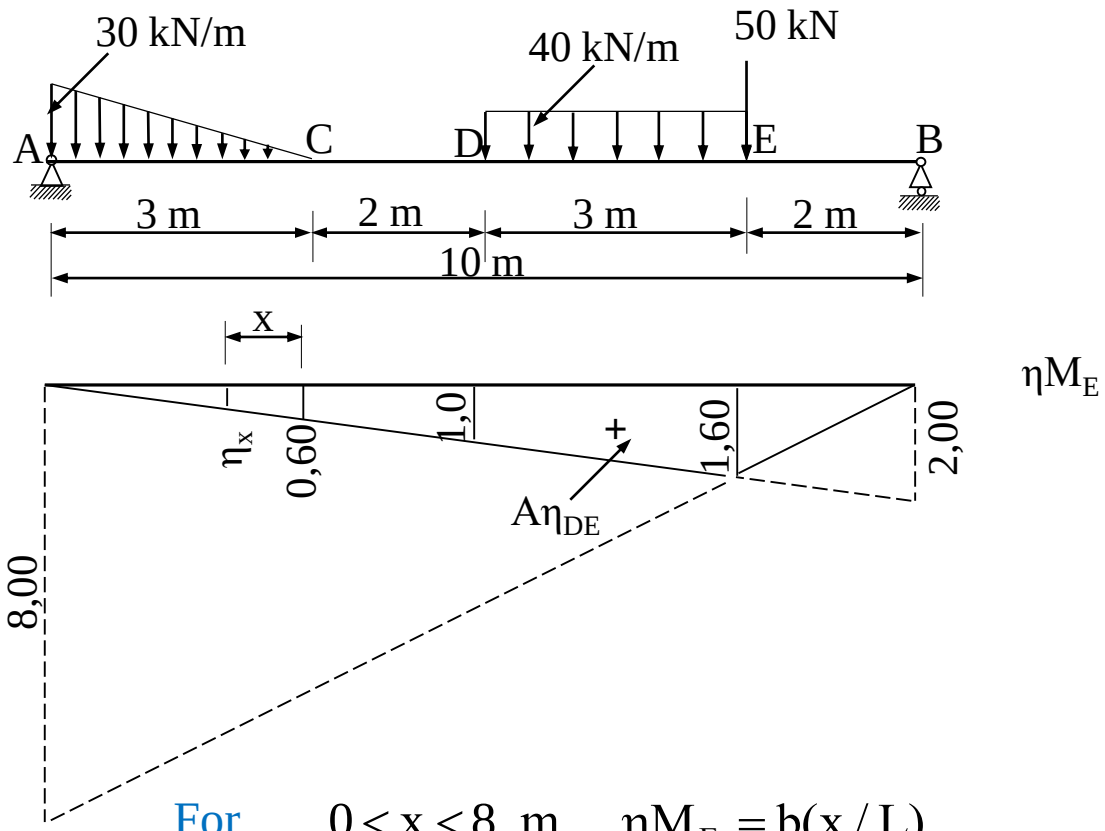
$$A\eta_{DE} = -1{,}95 \text{ m}$$

$$V_{EB} = -\int_0^3 10x \cdot (0{,}30 - 0{,}10x)dx - 40 \times 1{,}95 - 50 \times 0{,}80$$

$$V_{EB} = -[3x^2/2 - x^3/3]_0^3 - 78{,}0 - 40{,}0$$

$$V_{EB} = -4{,}5 - 78{,}0 - 40{,}0 = -122{,}50 \text{ kN}$$

e) M_E bending moment



For $0 < x < 8$ m $\quad \eta M_E = b(x/L)$

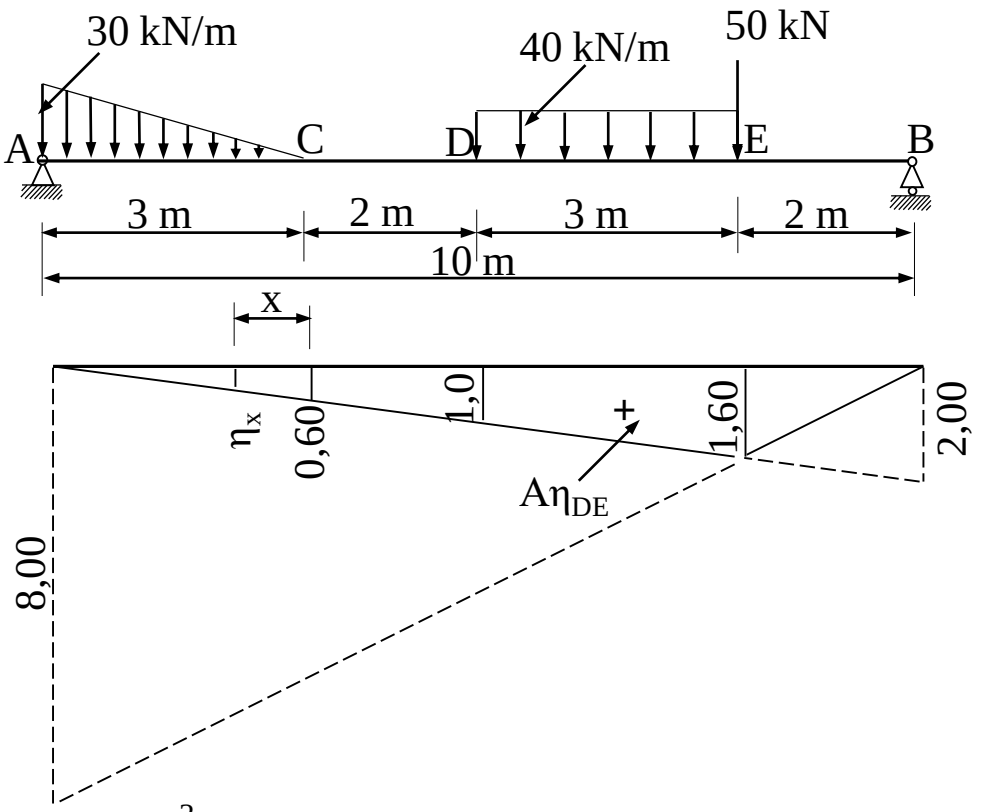
For 8 m $< x < 10$ m $\quad \eta M_E = a[(L-x)/L]$

$\eta_E = (2/10)\cdot 8{,}0 = 1{,}60$ m $\quad \eta_D = (5/10)\cdot 2{,}0 = 1{,}00$ m

$\eta_C = (3/10)\cdot 2{,}0 = 0{,}6$ m $\quad \eta_x = 0{,}6\cdot(3-x)/3 = 0{,}6 - 0{,}20x$

$q(x) = (x/3)\cdot 30 = 10x$

$A\eta_{DE} = 0{,}5\times(1{,}00+1{,}6)\times 3{,}0 = 3{,}90$ m^2



$$q(x) = 10x$$

$$\eta_x = 0{,}6 - 0{,}20x$$

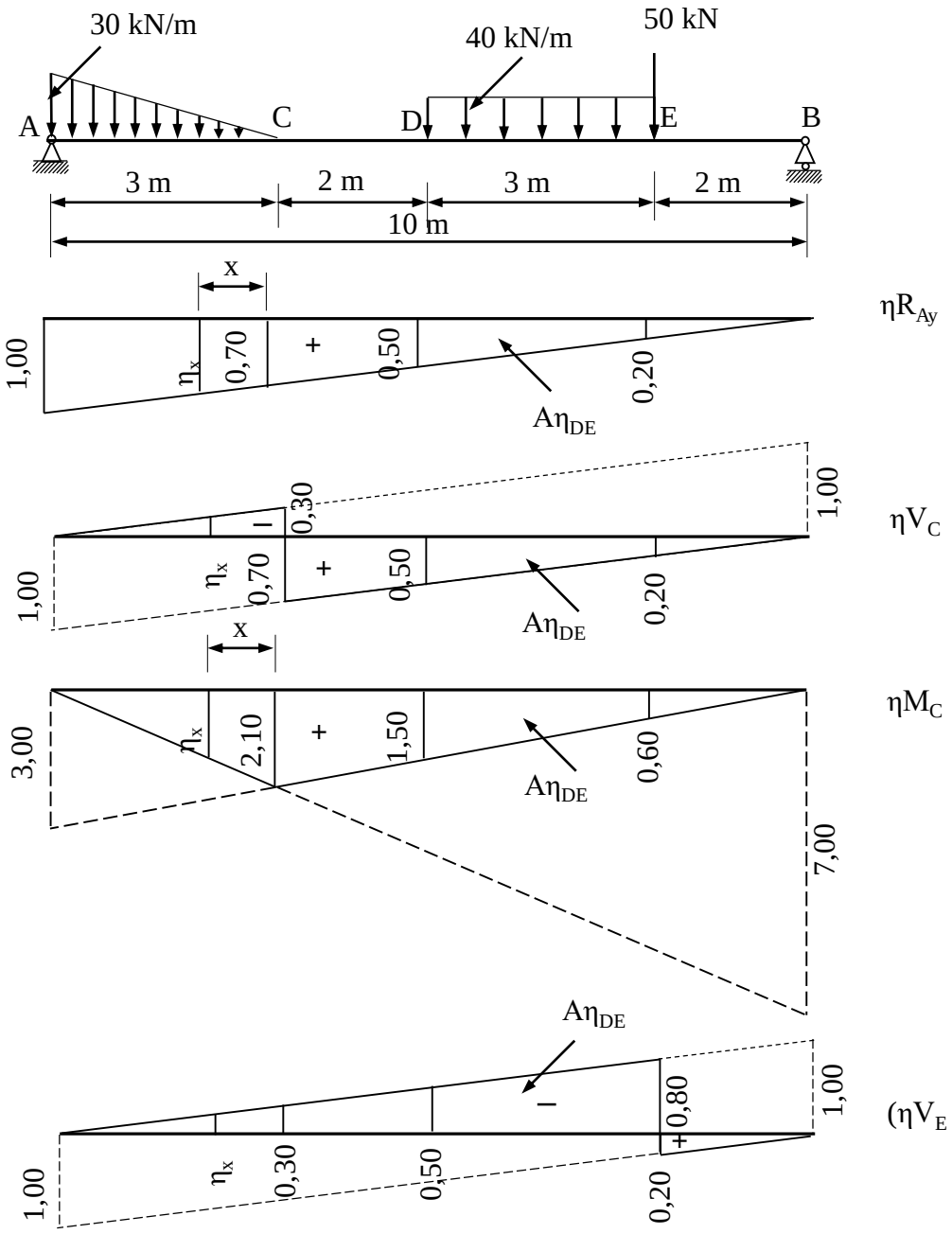
$$A\eta_{DE} = 3{,}90 \ \text{m}^2$$

$$M_E = \int_0^3 q(x) \cdot \eta_x \cdot dx + q \cdot A\eta_{DE} + Q \cdot \eta_E$$

$$M_E = \int_0^3 10x \cdot (0{,}6 - 0{,}20x) \cdot dx + 40 \times 3{,}90 + 50 \times 1{,}60$$

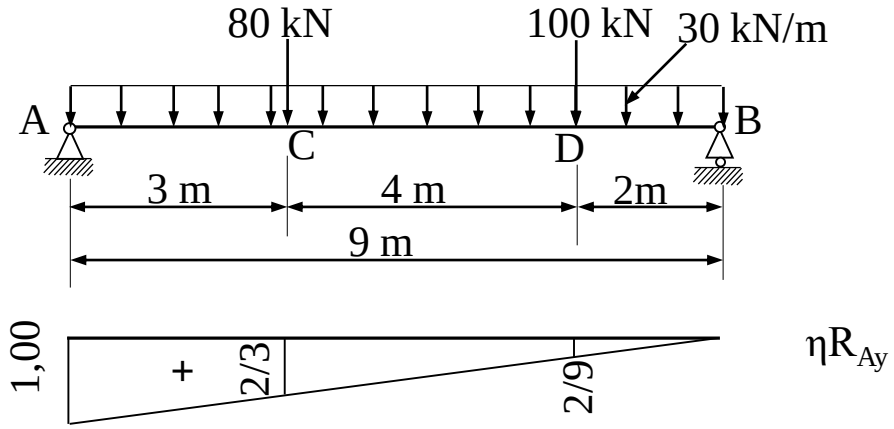
$$M_E = [6 \cdot x^2 / 2 - 2x^3 / 3]_0^3 + 156{,}0 + 80{,}0$$

$$M_E = 9{,}0 + 156{,}0 + 80{,}0 = 245{,}0 \ \text{kNm}$$


30 kN/m
40 kN/m
50 kN
A
C
D
E
B
3 m
2 m
3 m
2 m
10 m
x
ηR_Ay
1,00
η_x
0,70
+
0,50
0,20
Aη_DE
ηV_C
1,00
0,30
−
η_x
0,70
+
0,50
0,20
1,00
Aη_DE
x
ηM_C
3,00
η_x
2,10
+
1,50
0,60
Aη_DE
7,00
Aη_DE
(ηV_E
1,00
−
+0,80
1,00
η_x
0,30
0,50
0,20

Example 4)

Determine the vertical support reaction of A, shear force and moment at point C and D by using influence line method.



Solution:

a) R_{Ay} support reaction

$$\eta R_{Ay} = (L - x)/L$$

$$x = 0 \longrightarrow \eta_A = 1 \qquad x = L \longrightarrow \eta_B = 0$$

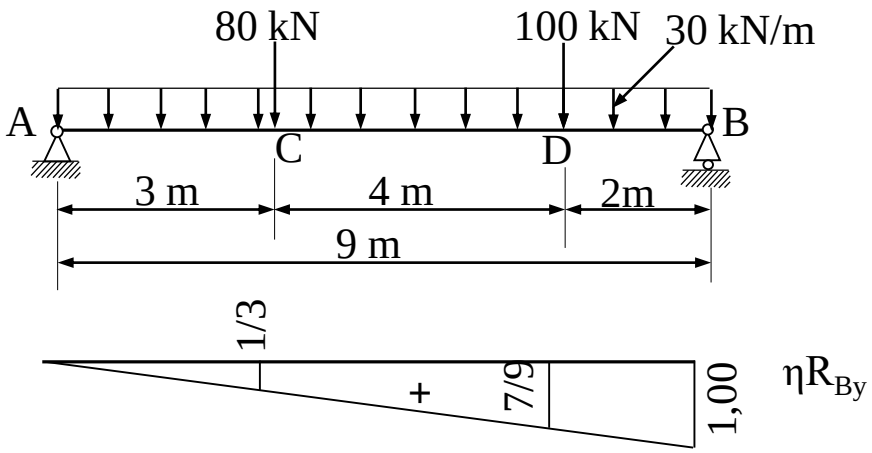
Influence line ordinates at the location of point loads

$$\eta_C = (6/9)\times 1{,}0 = 2/3 \qquad \eta_D = (2/9)\times 1{,}0 = 2/9$$

Influence line area under the uniform distributed load $A\eta = 0{,}5\times 1{,}0\times 9{,}0 = 4{,}5$ m

$$R_{Ay} = 80\times(2/3) + 100\times(2/9) + 30\times 4{,}50 = 210{,}556 \text{ kN}$$

b) R_{By} support reaction



R_{By} influence line function

$$\eta R_{By} = x / L$$

$$x = 0 \longrightarrow \eta_A = 0$$

$$x = L \longrightarrow \eta_B = 1$$

Influence line ordinates at the location of point loads

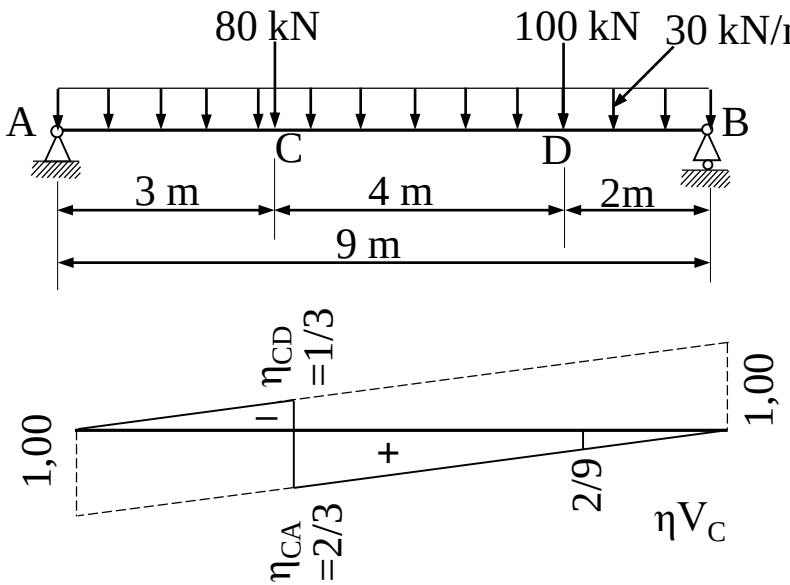
$$\eta_C = (3/9) \times 1{,}0 = 1/3 \qquad \eta_D = (7/9) \times 1{,}0 = 7/9$$

Influence line area under the uniform distributed load

$$A\eta = 0{,}5 \times 1{,}0 \times 9{,}0 = 4{,}5 \ \ \text{m}$$

$$R_{Ay} = 80 \times (1/3) + 100 \times (7/9) + 30 \times 4{,}50 = 239{,}444 \ \ \text{kN}$$

c) Shear force at point C



$0 < x < 3$ m $\quad \eta V_{CD} = -x / L$

3 m $< x < 9$ m

$\eta V_{CA} = (L - x) / L$

$\eta_{CA} = (6/9) \cdot 1{,}0 = 2/3$

$\eta_{CD} = -(3/9) \cdot 1{,}0 = -1/3$

$\eta_D = (2/9) \cdot 1{,}0 = 2/9$

Influence line area under the uniform distributed load

$$A\eta = -0{,}5 \times 3{,}0 \times (1/3) + 0{,}5 \times 6{,}0 \times (2/3) = 1{,}50 \text{ m}$$

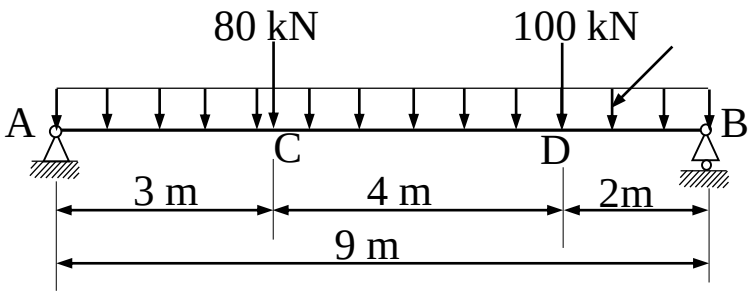
Point load at C creates sudden change on shear force diagram. V_{CA} and V_{CD} are estimated for left and right of point C.

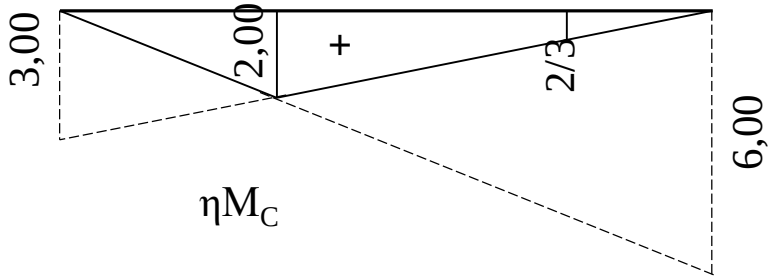
V_{CA} is estimated with the multiplication of η_{CA} and V_{CD} is estimated with the multiplication of η_{CD}.

$$V_{CA} = 80 \times (2/3) + 100 \times (2/9) + 30 \times 1{,}50 = 120{,}556 \text{ kN}$$

$$V_{CD} = -80 \times (1/3) + 100 \times (2/9) + 30 \times 1{,}50 = 40{,}556 \text{ kN}$$

c) M_C bending moment





$$0 < x < 3 \ \text{m} \longrightarrow \eta M_C = b(x/L)$$

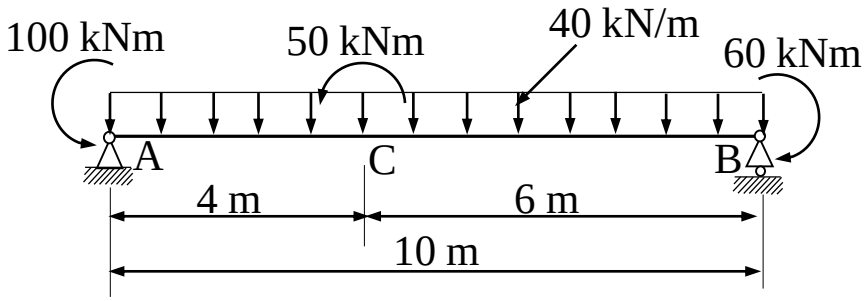
$$3 \ \text{m} < x < 9 \ \text{m} \longrightarrow \eta M_C = a[(L-x)/L]$$

$$\eta_C = (3/9)\cdot 6{,}0 = 2{,}0 \ \text{m} \qquad \eta_D = (2/9)\cdot 3{,}0 = 2/3 \ \text{m}$$

$$A\eta = 0{,}5 \times 2{,}0 \times 9{,}0 = 9{,}0 \ \text{m}^2$$

$$M_C = 80 \times 2{,}0 + 100 \times (2/3) + 30 \times 9{,}0 = 496{,}67 \ \text{kNm}$$

Example 5)

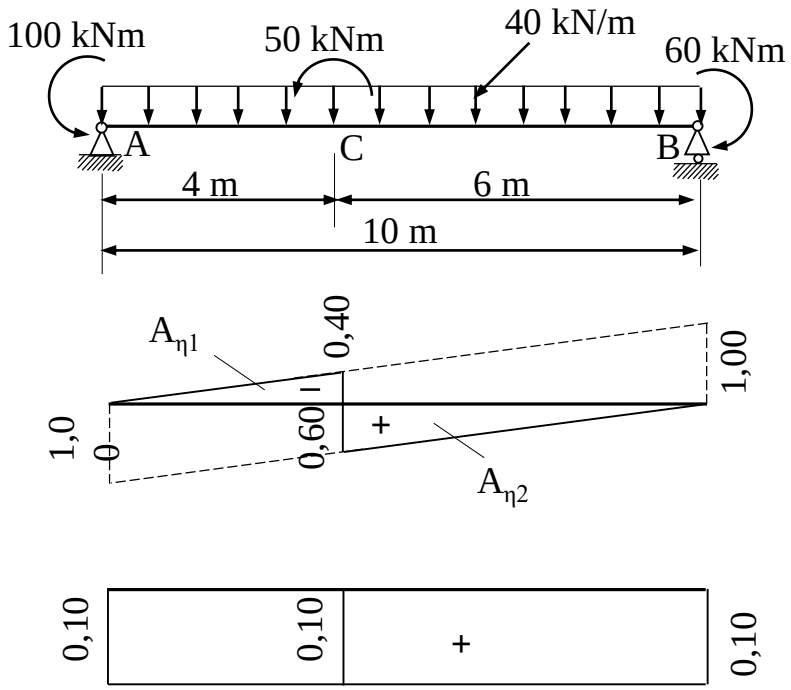


Determine the shear force and moment at point C by using influence line method.

Solution:

Since there are both a vertical load and point moment, influence lines for point load and point moment are drawn separately.

a) Shear force at point C



- V_C influence line for unit point load

$0 < x < 4$ m $\quad \eta V_C = -x / L$

4 m $< x < 10$ m

$\eta V_C = (L - x) / L$

$\eta_{CA} = 0{,}60 \quad \eta_{CB} = 0{,}40$

Area between AC

$A\eta_1 = -0{,}5 \times 0{,}40 \times 4{,}0 = -0{,}80$ m

Area between AC

$A\eta_2 = 0{,}5 \times 0{,}60 \times 6{,}0 = 1{,}80$ m

- V_C influence line for unit point moment

$0 < x < 4$ m $\quad \eta V_C = 1/L$ $\quad 4$ m $< x < 10$ m $\quad \eta V_{CC} = 1/L$

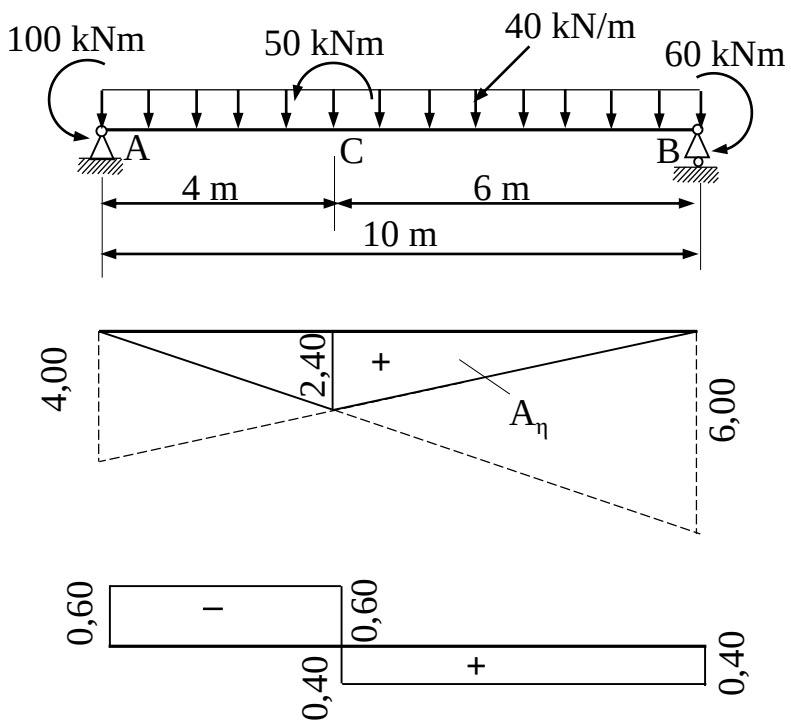
Ordinates under the point moments:

$\eta_A = 0{,}10 \quad \eta_B = 0{,}10 \quad \eta_C = 0{,}10$

Note: Since the 60 kNm moment rotation is in clockwise, the sign of 60 kNm multiplication is (-),

$V_C = 40 \times (1{,}80 - 0{,}80) + 100 \times 0{,}10 + 50 \times 0{,}10 + (-60) \times 0{,}1 = 49{,}00$ kN

b) Bending moment at point C



- M_C influence line for unit point load

$0 < x < 4 \ \text{m} \qquad \eta M_{CB} = b(x/L)$

$4 \ \text{m} < x < 10 \ \text{m}$

$\eta M_{CA} = a[(L-x)/L]$

$\eta_C = 4{,}0 \times 6{,}0 / 10 = 2{,}4 \ \text{m}$

$A\eta = 0{,}5 \times 2{,}4 \times 10{,}0 = 12{,}00 \ \text{m}^2$

- M_C influence line for unit point moment

$0 < x < 4 \ \text{m} \qquad \eta M_{CB} = -b/L$

$4 \ \text{m} < x < 10 \ \text{m} \qquad \eta M_{CA} = a/L$

Ordinates under the point moments:

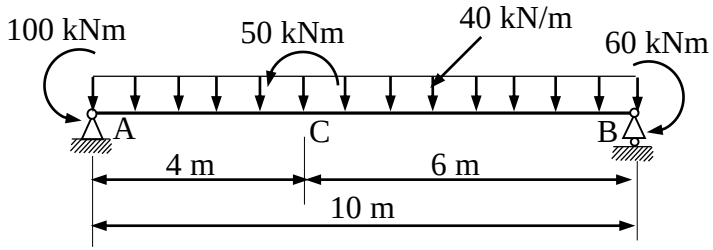
$\eta_A = -0{,}60 \qquad \eta_B = 0{,}40 \qquad \eta_{CA} = 0{,}40 \qquad \eta_{CB} = -0{,}60$

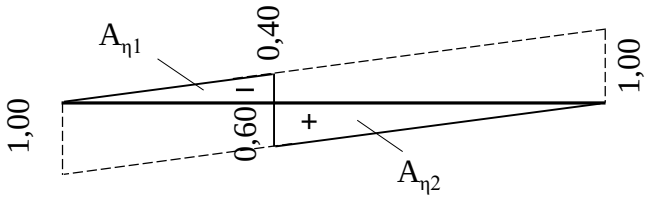
Since point moment creates sudden change on moment diagram, moments at the right and left of the point C are obtained

$M_{CA} = 40 \times 12{,}0 + 100 \times (-0{,}60) + 50 \times 0{,}40 + (-60) \times 0{,}4 = \ 416{,}00 \ \text{kNm}$

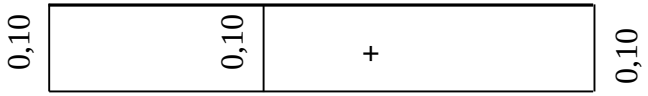
$M_{CB} = 40 \times 12{,}0 + 100 \times (-0{,}60) + 50 \times (-0{,}60) + (-60) \times 0{,}4$

$M_{CB} = 366{,}00 \ \text{kNm}$

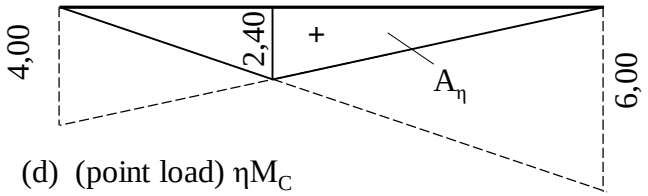




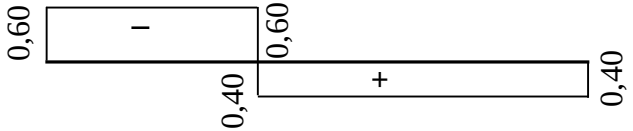
(b) (point load) ηV_C



(c) (point moment) ηV_C



(d) (point load) ηM_C



(e) (point moment) ηM_C Şekil 5.20

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

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